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"Investigation into a Disease" Project

Submission deadline: Friday, November 20, 2026, at 11:59 p.m. No late submissions will be accepted.

Your group will select a real-world dataset in the field of health or genomics, ask a specific question about a disease, define a pattern (a sequence, a variant, a combination of criteria, or an expression profile), and build a **reproducible pipeline** that automatically searches for it.

The topic is open-ended. The only requirement is the technical framework: it must cover the five chapters of the module (shell, file manipulation, system administration, scripting, Python).

Here are a few ideas (you may suggest others):

- antibiotic resistance genes: bacterial genomes from NCBI, a pattern present in resistant strains and absent in others;
- variants and genetic disease: ClinVar and 1000 Genomes for a single gene or a panel, common pathogenic variants;
- expression in cancer: GEO dataset comparing tumor to healthy tissue, genes that distinguish the two groups;
- risk factors: UCI or Kaggle datasets (diabetes, heart disease), a combination of criteria that predicts the disease;
- Virus evolution: NCBI Virus, mutations that have emerged by date or region.

Data constraints: public and freely downloadable; between 100 MB and 1 GB; no non-anonymized personal data or datasets with restricted access. **One topic and one dataset per group**: in the event of a duplicate, the first group to submit its form takes priority.

3. Topic Validation Form

Submit via Google Classroom by Friday, October 23, 2026, at 11:59 PM. One page, including the following sections:

Section	Expected Content
Group and Members	Group number, names of the 5 students
Dataset	Name, ID (or link), size, download date
Disease and question	The question asked, in one or two sentences

You will receive a response marked “approved” or “needs revision” within 3 days. Do not begin writing the paper until it has been approved.

4. Required technical framework

Your submission must contain a single **run.sh** script that executes the entire pipeline, from raw data to results.

- **Shell:** data acquisition, filtering, and cleaning using standard tools (`grep`, `sed`, `awk`, `cut`, `sort`, `uniq`...), by chaining commands via pipes.
- **File handling:** clear directory structure (`data/`, `scripts/`, `results/`, `logs/`), checksums, archiving of results.
- **System administration:** isolated and reproducible software environment, scheduled tasks, timestamped logs, proper access permissions.
- **Scripting:** arguments (`getopts`), `-h` help, error and signal handling (`trap`), return codes.
- **Python:** statistical analysis of results and figures, called from `run.sh`.

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Your pipeline must run there without manual intervention, following only your README.md.

5. To be submitted on Google Classroom

An archive named Projet_Maladie_Groupe_XX.zip containing:

- LIEN_GIT.txt: address of the team repository **and hash of the final commit** (only this commit will be evaluated);
- the pipeline code (run.sh and scripts);
- README.md: installation instructions, user guide, work breakdown, and description of the output files;
- SOURCES.md: identifier, URL, and download date of the dataset;
- report.pdf: maximum 4 pages (problem statement, data, method and definition of the pattern, results, limitations). The data itself should not be included: your script must download it and verify its integrity.

A working version must be submitted by 11:59 PM on Monday, November 9, 2026, for the cross-validation test described in point 7.

Failure to submit the archive or submission after the deadline will result in a score of 0.

6. Technical Criteria Grid

Each criterion is evaluated as "yes" or "no," without subjective judgment. There are 20 criteria, worth 1 point each.

Chapter	Criteria
Shell	run.sh displays help when run with -h · at least two options processed with getopts · a missing file results in a non-zero exit code and a clear error message · at least one sequence of 3 or more commands piped through the data
Files	directory structure described in the README · checksums calculated and verified by the script · results archived in a dated archive (tar) · complete SOURCES.md
System Administration	Reproducible environment provided (environment.yml or requirements.txt) · Cron job or timer provided and documented · Timestamped log in logs/ · Executable scripts and read-only raw data
Script	trap that deletes temporary files upon termination · exits on the first error (set -euo pipefail) with a clear message · at least 3 reused functions · can be restarted without manual cleanup
Python	script called by run.sh using argparse · at least one automatically generated figure · numerical results written to a CSV or JSON file · at least one automated test of the processing

7. Cross-group testing

From November 9 to 13, each group receives a working version from another group and attempts to run it **using only its README**. Within 48 hours, the group submits a short report (template provided): does the pipeline start, does it complete, does it produce the expected files, and what is missing from the README? The feedback is used to make corrections before the final submission.

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8. Evaluation

Element	Part
Pipeline execution on a clean machine (it starts, completes, and produces the files listed in the README)	20%
Technical Criteria Grid (Item 6)	25%
4-page report	20%
Cross-check: report submitted and relevant	5%
Individual oral defense	30%

9. Oral defense

- 25 minutes per group, including a 10-minute presentation covering the key points.
- The remainder of the time: **specific individual questions** on the concepts used and **minor live code modifications**, requested from the designated student.
- Grades within the same group are not necessarily identical. The

dates of the presentations will be announced after the project is

submitted.

10. Schedule

Date	Stage
Friday, October 23	Topic Approval Form
October 26–November 6	Development, 5-minute check-in with the instructor by group
Monday, November 9	Functional version submitted
November 9–13	Cross-testing
Friday, November 20, 11:59 p.m.	Final submission

11. Rules

You are responsible for every line you submit: you must be able to explain and modify it orally, regardless of how it was produced. Projects that are very similar across groups will be reviewed. If a student disputes their grade, they have 48 hours after it is posted to write to the instructor explaining their case.

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